

# **INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY**

## **Week 4 Practical Laboratory Linux Security Monitoring and Auditing**

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## Module 1 System Auditing with auditd

### Activity 1.1 Verify or Install auditd

- Installation of auditd was successful.

```
Setting up auditd (1:4.12-1+b2)...\nupdate-rc.d: We have no instructions for the auditd init script.\nupdate-rc.d: It looks like a non-network service, we enable it.\naudit-rules.service is a disabled or a static unit, not starting it.\nauditd.service is a disabled or a static unit, not starting it.\nSetting up audispd-plugins (1:4.1.2-1+b2)...\nProcessing triggers for libc-bin (2.42-13)...\nProcessing triggers for man-db (2.13.1-1)...\nProcessing triggers for kali-menu (2026.1.3)...\n\n(kali@kali)-[~]\n$ sudo systemctl start auditd\n\n(kali@kali)-[~]\n$ sudo systemctl enable auditd\nSynchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.\nExecuting: /usr/lib/systemd/systemd-sysv-install enable auditd\nCreated symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' -> '/usr/lib/systemd/system/auditd.service'.\n\n(kali@kali)-[~]\n$
```

### Activity 1.2 Configure Audit Rules

- Creating dedicated files and restart the audit service

```
(kali@kali)-[~]\n$ sudo nano /etc/audit/rules.d/custom.rules\n[sudo] password for kali:\n\n(kali@kali)-[~]\n$ sudo systemctl restart auditd\n\n(kali@kali)-[~]\n$ sudo auditctl -l\n-w /etc/passwd -p rwx -k passwd_changes\n-w /etc/shadow -p rwx -k shadow_changes\n-a always,exit -F arch=b64 -S execve -F key=program_execution\n-a always,exit -F arch=b32 -S execve -F key=program_execution\n-w /var/log/auth.log -p wa -k auth_failures\n\n(kali@kali)-[~]\n$
```

- ausearch result for a configured audit key

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
(kali㉿kali)-[~]  
$ sudo aureport  
  
Summary Report  
=====
```

|                                                  |                                                   |
|--------------------------------------------------|---------------------------------------------------|
| Range of time in logs:                           | 10/02/2026 05:23:25.724 - 10/02/2026 05:58:52.493 |
| Selected time for report:                        | 10/02/2026 05:23:25 - 10/02/2026 05:58:52.493     |
| Number of changes in configuration:              | 11                                                |
| Number of changes to accounts, groups, or roles: | 0                                                 |
| Number of logins:                                | 0                                                 |
| Number of failed logins:                         | 0                                                 |
| Number of authentications:                       | 1                                                 |
| Number of failed authentications:                | 0                                                 |
| Number of users:                                 | 3                                                 |
| Number of terminals:                             | 7                                                 |
| Number of host names:                            | 2                                                 |
| Number of executables:                           | 14                                                |
| Number of commands:                              | 15                                                |
| Number of files:                                 | 17                                                |
| Number of AVC's:                                 | 0                                                 |
| Number of MAC events:                            | 0                                                 |
| Number of failed syscalls:                       | 2                                                 |
| Number of anomaly events:                        | 0                                                 |
| Number of responses to anomaly events:           | 0                                                 |
| Number of crypto events:                         | 0                                                 |
| Number of integrity events:                      | 0                                                 |
| Number of virt events:                           | 0                                                 |
| Number of keys:                                  | 4                                                 |
| Number of process IDs:                           | 47                                                |
| Number of events:                                | 345                                               |

```
(kali㉿kali)-[~]  
$ █
```

Interpretation: The service status should confirm whether auditd is active. An active audit service provides the foundation for collecting security-relevant audit evidence. The laboratory specifically requires evidence showing whether auditd is running.

## Module 2 Log Management and Analysis

```
Oct 02 08:37:31 kali NetworkManager[808]: <info> [1790944651.2685] manager: NetworkManager state is now CONNECTING
Oct 02 08:37:31 kali NetworkManager[808]: <info> [1790944651.2688] device (eth0): state change: prepare → config (reason '
Oct 02 08:37:31 kali NetworkManager[808]: <info> [1790944651.2767] device (eth0): state change: config → ip-config (reason '
Oct 02 08:37:31 kali dbus-daemon[623]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='d
Oct 02 08:37:31 kali NetworkManager[808]: <info> [1790944651.3708] dhcp4 (eth0): activation: beginning transaction (timeout
Oct 02 08:37:31 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
Oct 02 08:37:32 kali dbus-daemon[623]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
Oct 02 08:37:32 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
Oct 02 08:37:32 kali NetworkManager[808]: <info> [1790944652.4736] dhcp4 (eth0): state changed new lease, address=192.168.1
Oct 02 08:37:33 kali NetworkManager[808]: <info> [1790944653.1561] dhcp4 (eth0): state changed new lease, address=192.168.1
Oct 02 08:37:33 kali NetworkManager[808]: <info> [1790944653.1717] policy: set 'Wired connection 1' (eth0) as default for I
Oct 02 08:37:34 kali NetworkManager[808]: <info> [1790944654.7426] device (eth0): state change: ip-config → ip-check (reas
Oct 02 08:37:35 kali NetworkManager[808]: <info> [1790944655.3739] device (eth0): state change: ip-check → secondaries (re
Oct 02 08:37:35 kali NetworkManager[808]: <info> [1790944655.3744] device (eth0): state change: secondaries → activated (r
Oct 02 08:37:35 kali NetworkManager[808]: <info> [1790944655.3751] manager: NetworkManager state is now CONNECTED_SITE
Oct 02 08:37:35 kali NetworkManager[808]: <info> [1790944655.3757] device (eth0): Activation: successful, device activated.
Oct 02 08:37:35 kali NetworkManager[808]: <info> [1790944655.3778] manager: NetworkManager state is now CONNECTED_GLOBAL
Oct 02 08:37:42 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Oct 02 08:39:01 kali CRON[6866]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 08:39:01 kali CRON[6868]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /us
Oct 02 08:39:01 kali CRON[6866]: pam_unix(cron:session): session closed for user root
lines 28772-28805/28805 (END)
```

```
(kali㉿kali)-[~]
$ sudo journalctl -f
Oct 02 08:44:42 kali sudo[6983]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 02 08:45:01 kali CRON[6988]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 08:45:01 kali CRON[6990]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Oct 02 08:45:01 kali CRON[6988]: pam_unix(cron:session): session closed for user root
Oct 02 08:45:18 kali sudo[6983]: pam_unix(sudo:session): session closed for user root
Oct 02 08:46:25 kali sudo[7002]: kali : TTY=pts/3 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -p err
Oct 02 08:46:25 kali sudo[7002]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 02 08:46:32 kali sudo[7002]: pam_unix(sudo:session): session closed for user root
Oct 02 08:46:54 kali sudo[7012]: kali : TTY=pts/3 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -f
Oct 02 08:46:54 kali sudo[7012]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
```

```
(kali㉿kali)-[~]
$ sudo less /var/log/auth.log
/var/log/auth.log: No such file or directory

(kali㉿kali)-[~]
$
```

- These rules monitor sensitive account files, program execution and authentication-log changes.

## Module 3 Linux Security Assessment with Lynis

```

(kali㉿kali)-[~]
$ sudo apt install lynis
Installing:
  lynis

Installing dependencies:
  menu

Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2077
  Download size: 631 kB
  Space needed: 3,293 kB / 58.3 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 lynis all 3.1.6-1 [276 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Fetched 631 kB in 15s (42.5 kB/s)
Selecting previously unselected package lynis.
(Reading database... 439286 files and directories currently installed.)

```

To return to your computer, move the mouse pointer outside or press Ctl+Alt

```

Lynis 3.1.6

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2025, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all setti

(kali㉿kali)-[~]
$ sudo ./lynis audit system
sudo: ./lynis: command not found

```

Lynis as an auditing tool that checks Unix-like systems and reports warnings, suggestions and a hardening index. It is an assessment tool rather than an active penetration-testing tool.

#### Module 4 SIEM, Automation and Continuous Monitoring

| Control / Objective | Evidence | Owner | Observed Status | KPI/KRI or Threshold | Risk / Significance | Remediation | Retest / Follow-up |
|---------------------|----------|-------|-----------------|----------------------|---------------------|-------------|--------------------|
|                     |          |       |                 |                      |                     |             |                    |
|                     |          |       |                 |                      |                     |             |                    |
|                     |          |       |                 |                      |                     |             |                    |
|                     |          |       |                 |                      |                     |             |                    |

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

### Governance Escalation Questions

The highest-priority finding should be selected from the actual evidence after considering severity, exposure, likelihood, business impact and whether the issue affects a critical service. I would escalate a high/critical security condition to the CISO or appropriate management committee when it exceeds the approved risk threshold, affects a critical system, remains unresolved beyond its SLA, or indicates a control failure.

Remediation ownership should be assigned to the role responsible for operating the affected control.